1. Defining Problem Statement and Analysing basic metrics

In [ ]:
```
import pandas as pd
import numpy as np
import matplotlib
import matplotlib.pyplot as plt
import seaborn as sns
```

In [ ]:
```
netflix_df = pd.read_csv('netflix.csv')
```

In [ ]:
```
netflix_df.shape
```

Out[ ]:
```
(8154, 12)
```

In [ ]:
```
netflix_df.head()
```

Out[ ]:

	show_id	type	title	director	cast	country	date_added	release_year	rating	duration	listed_in	description
0	s1	Movie	Dick Johnson Is Dead	Kirsten Johnson	NaN	United States	September 25, 2021	2020	PG-13	90 min	Documentaries	As her father nears the end of his life, filmm...
1	s2	TV Show	Blood & Water	NaN	Ama Qamata, Khosi Ngema, Gail Mabalane, Thaban...	South Africa	September 24, 2021	2021	TV-MA	2 Seasons	International TV Shows, TV Dramas, TV Mysteries	After crossing paths at a party, a Cape Town t...
2	s3	TV Show	Ganglands	Julien Leclercq	Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi...	NaN	September 24, 2021	2021	TV-MA	1 Season	Crime TV Shows, International TV Shows, TV Act...	To protect his family from a powerful drug lor...
3	s4	TV Show	Jailbirds New Orleans	NaN	NaN	NaN	September 24, 2021	2021	TV-MA	1 Season	Docuseries, Reality TV	Feuds, flirtations and toilet talk go down amo...
4	s5	TV Show	Kota Factory	NaN	Mayur More, Jitendra Kumar, Ranjan Raj, Alam K...	India	September 24, 2021	2021	TV-MA	2 Seasons	International TV Shows, Romantic TV Shows, TV ...	In a city of coaching centers known to train I...

The dataset contains over 8807 titles, 12 descriptions. It looks like a movie/TV shows data frame but ratings are not available. We can also see that there are NaN values in some columns.

2. Observations on the shape of data, data types of all the attributes, conversion of categorical attributes to 'category' (If required), missing value detection, statistical summary

In [ ]:
```
#attributes
netflix_df.columns
```

Out[ ]:
```
Index(['show_id', 'type', 'title', 'director', 'cast', 'country', 'date_added',
       'release_year', 'rating', 'duration', 'listed_in', 'description'],
      dtype='object')
```

In [ ]:
```
#datatypes of all attributes
netflix_df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 8154 entries, 0 to 8153
Data columns (total 12 columns):
 #   Column        Non-Null Count  Dtype 
---  ------        --------------  ----- 
 0   show_id       8154 non-null   object
 1   type          8154 non-null   object
 2   title         8154 non-null   object
 3   director      5636 non-null   object
 4   cast          7407 non-null   object
 5   country       7351 non-null   object
 6   date_added    8145 non-null   object
 7   release_year  8154 non-null   int64 
 8   rating        8150 non-null   object
 9   duration      8150 non-null   object
 10  listed_in     8153 non-null   object
 11  description   8153 non-null   object
dtypes: int64(1), object(11)
memory usage: 764.6+ KB
```

In [ ]:
```
#statistical summary
netflix_df.describe()
```

Out[ ]:

	release_year
count	8154.000000
mean	2014.503066
std	8.411897
min	1925.000000
25%	2014.000000
50%	2017.000000
75%	2019.000000
max	2021.000000

In [ ]:
```
#missing value detection
print('\nColumns with missing value:')
print(netflix_df.isnull().any())
```

```
Columns with missing value:
show_id         False
type            False
title           False
director         True
cast             True
country          True
date_added       True
release_year    False
rating           True
duration         True
listed_in        True
description      True
dtype: bool
```

There are a few columns that contain null values, “director,” “cast,” “country,” “date_added,” “rating.”

In [ ]:
```
netflix_df.T.apply(lambda x: x.isnull().sum(), axis = 1)
```

Out[ ]:

	0
show_id	0
type	0
title	0
director	2518
cast	747
country	803
date_added	9
release_year	0
rating	4
duration	4
listed_in	1
description	1

dtype: int64

In [ ]:
```
netflix_df.isnull().sum().sum()
```

Out[ ]: 4087

There are a total of 4307 null values in the dataset.

In [9]:
```
netflix_df['director'] = netflix_df['director'].fillna("No Director")
netflix_df['cast'] = netflix_df['cast'].fillna("No Cast")
netflix_df['country'] = netflix_df['country'].fillna("Country Unavailable")
netflix_df.dropna(subset=["date_added", "rating", "duration"], inplace=True)
```

In [ ]:
```
netflix_df["date_added"] = pd.to_datetime(netflix_df["date_added"].str.strip())
netflix_df["year_added"] = netflix_df["date_added"].dt.year
netflix_df["month_added"] = netflix_df["date_added"].dt.month_name()


netflix_df["listed_in"] = netflix_df["listed_in"].str.split(", ")

netflix_df["country"] = netflix_df["country"].str.strip()
```

In [ ]:
```
netflix_df.isnull().any()
```

Out[ ]:

	0
show_id	False
type	False
title	False
director	False
cast	False
country	False
date_added	False
release_year	False
rating	False
duration	False
listed_in	False
description	False
year_added	False
month_added	False

dtype: bool

In [ ]:

```
#statistical summary after cleaning
netflix_df.describe()
```

Out[ ]:

	date_added	release_year	year_added
count	8137	8137.000000	8137.000000
mean	2019-06-08 12:31:03.487771904	2014.507189	2018.930073
min	2008-01-01 00:00:00	1925.000000	2008.000000
25%	2018-05-01 00:00:00	2014.000000	2018.000000
50%	2019-08-09 00:00:00	2017.000000	2019.000000
75%	2020-09-17 00:00:00	2019.000000	2020.000000
max	2021-09-25 00:00:00	2021.000000	2021.000000
std	NaN	8.417845	1.585455

3. Non-Graphical Analysis: Value counts and unique attributes

In [ ]:
```
netflix_df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Index: 8137 entries, 0 to 8152
Data columns (total 14 columns):
 #   Column        Non-Null Count  Dtype
---  ------        --------------  -----
 0   show_id       8137 non-null   object
 1   type          8137 non-null   object
 2   title         8137 non-null   object
 3   director      8137 non-null   object
 4   cast          8137 non-null   object
 5   country       8137 non-null   object
 6   date_added    8137 non-null   datetime64[ns]
 7   release_year  8137 non-null   int64
 8   rating        8137 non-null   object
 9   duration      8137 non-null   object
 10  listed_in     8137 non-null   object
 11  description   8137 non-null   object
 12  year_added    8137 non-null   int32
 13  month_added   8137 non-null   object
dtypes: datetime64[ns](1), int32(1), int64(1), object(11)
memory usage: 921.8+ KB
```

In [ ]:
```
netflix_df.describe()
```

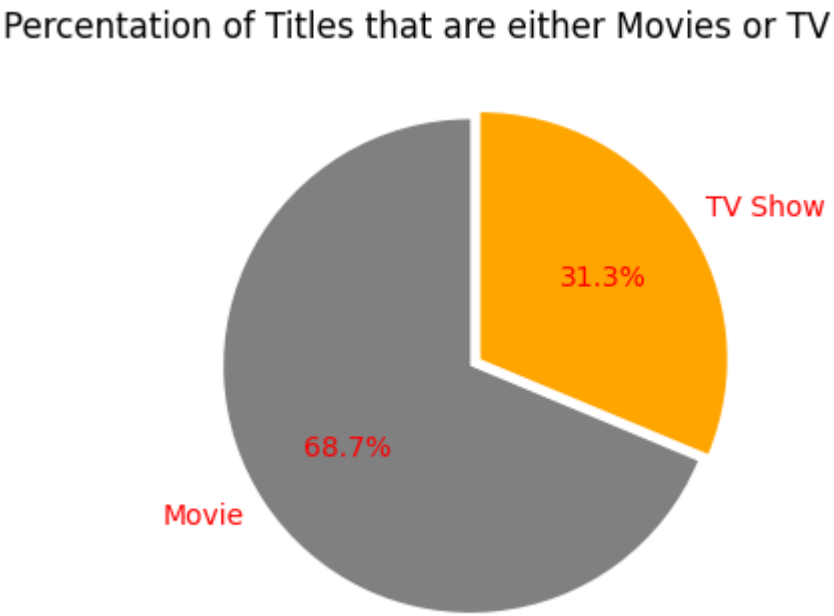
Out[ ]:

	date_added	release_year	year_added
count	8137	8137.000000	8137.000000
mean	2019-06-08 12:31:03.487771904	2014.507189	2018.930073
min	2008-01-01 00:00:00	1925.000000	2008.000000
25%	2018-05-01 00:00:00	2014.000000	2018.000000
50%	2019-08-09 00:00:00	2017.000000	2019.000000
75%	2020-09-17 00:00:00	2019.000000	2020.000000
max	2021-09-25 00:00:00	2021.000000	2021.000000
std	NaN	8.417845	1.585455

4. Visual Analysis - Univariate, Bivariate after pre-processing of the data

4.1 For continuous variable(s): Distplot, countplot, barplot, histogram for univariate analysis

In [ ]:
```
#percentage of movies and shows
plt.figure(figsize=(4,4))
plt.title("Percentation of Titles that are either Movies or TV Shows")
g=plt.pie(netflix_df.type.value_counts(),explode=(0.025,0.025),
labels=netflix_df.type.value_counts().index, colors=['grey','orange'],autopct='%1.1f%%',
startangle=90, textprops={"color":"red"})
plt.show()
```



Major content on Netflix comprises of Movies at 68.7% and TV shows at 31.3%.

In [ ]:
```
netflix_df
```

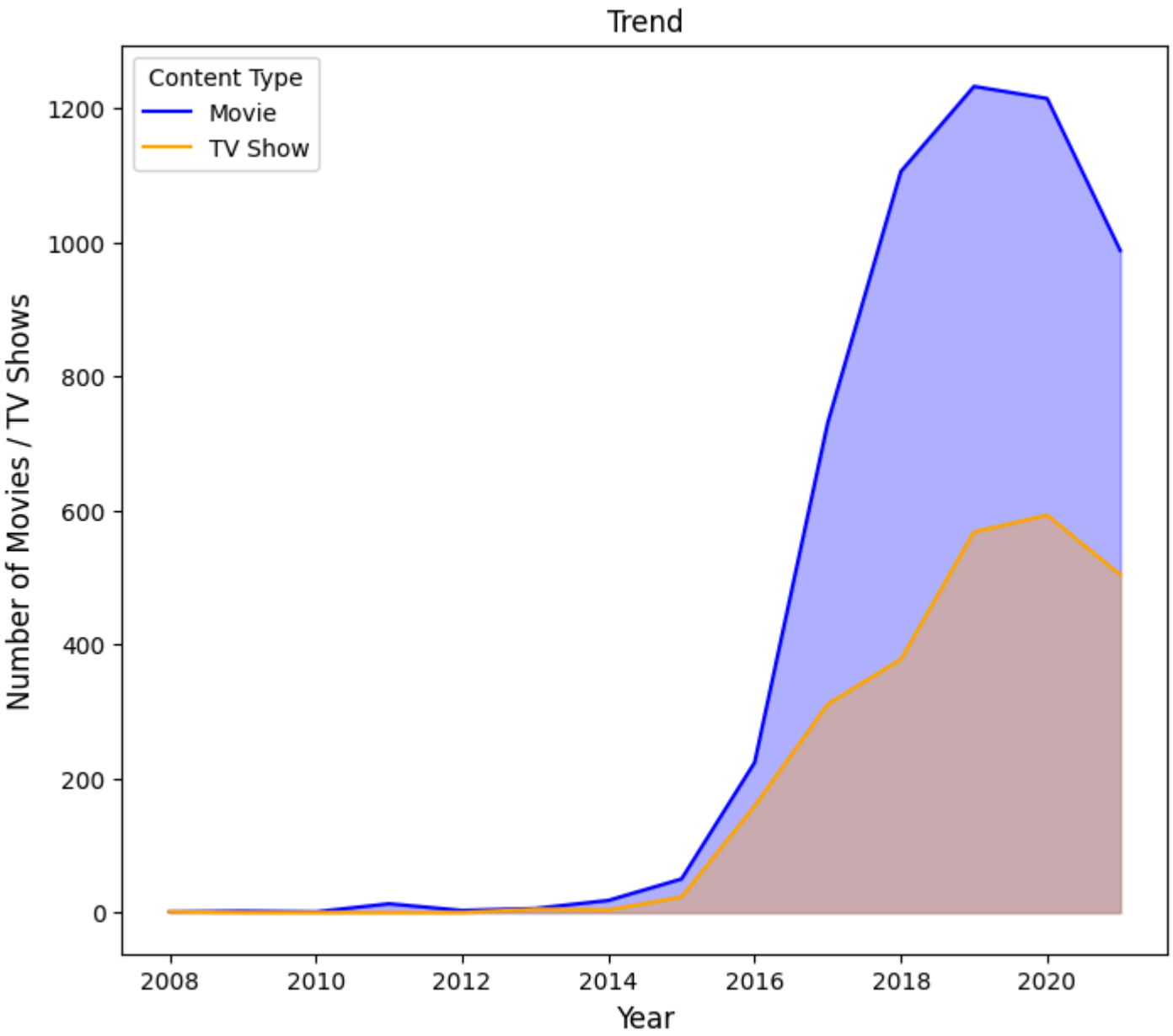
Out[ ]:

	show_id	type	title	director	cast	country	date_added	release_year	rating	duration	listed_in	descript
0	s1	Movie	Dick Johnson Is Dead	Kirsten Johnson	No Cast	United States	2021-09-25	2020	PG-13	90 min	[Documentaries]	As father ne the enc his film
1	s2	TV Show	Blood & Water	No Director	Ama Qamata, Khosi Ngema, Gail Mabalane, Thaban...	South Africa	2021-09-24	2021	TV-MA	2 Seasons	[International TV Shows, TV Dramas, TV Mysteries]	A cross paths part Cape To
2	s3	TV Show	Ganglands	Julien Leclercq	Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi...	Country Unavailable	2021-09-24	2021	TV-MA	1 Season	[Crime TV Shows, International TV Shows, TV Ac...	To prot his far fro powe drug l
3	s4	TV Show	Jailbirds New Orleans	No Director	No Cast	Country Unavailable	2021-09-24	2021	TV-MA	1 Season	[Docuseries, Reality TV]	Feu flirtati and to talk down am
4	s5	TV Show	Kota Factory	No Director	Mayur More, Jitendra Kumar, Ranjan Raj, Alam K...	India	2021-09-24	2021	TV-MA	2 Seasons	[International TV Shows, Romantic TV Shows, TV...	In a city coach cent known trai
...	...	...	...	...	...	...	...	...	...	...	...	

	show_id	type	title	director	cast	country	date_added	release_year	rating	duration	listed_in	descript
8148	s8149	TV Show	Tales of the City (1993)	No Director	Olympia Dukakis, Donald Moffat, Laura Linney, ...	United Kingdom, United States	2019-06-28	1993	TV-MA	1 Season	[TV Dramas]	In 19 naive M A Single moves Sa
8149	s8150	Movie	Tamanchey	Navneet Behal	Nikhil Dwivedi, Richa Chadha, Damandeep Singh ...	India	2017-10-15	2014	TV-MA	111 min	[Action & Adventure, International Movies, Rom...	A man a a woman the from law ar
8150	s8151	Movie	Tarif de nuit	Julien Seri	Jonathan Howard, Jonathan Demurger, Fanny Vale...	France	2018-03-01	2015	TV-MA	81 min	[International Movies, Thrillers]	A night in P quic unrav when tw
8151	s8152	Movie	Tarzan	Chris Buck, Kevin Lima	Tony Goldwyn, Minnie Driver, Glenn Close, Bria...	United States	2018-06-23	1999	G	89 min	[Children & Family Movies]	After be shipwrecl off Afri coas
8152	s8153	Movie	Tarzan 2	Brian Smith	Harrison Chad, George Carlin, Brad Garrett, Ro...	United States	2017-06-23	2005	G	72 min	[Children & Family Movies]	How did King of Jun ascenc the

8137 rows × 14 columns

In [ ]:
```
#year vs total amount of different content released.
per_year = netflix_df.groupby(["year_added", "type"]).size().unstack(fill_value=0)


years = per_year.index
movies = per_year["Movie"]
tv_shows = per_year["TV Show"]


plt.figure(figsize=(8,7))
sns.lineplot(x=years, y=movies, label="Movie", color="blue")
sns.lineplot(x=years, y=tv_shows, label="TV Show", color="orange")


plt.fill_between(years, movies, color="blue", alpha=0.3)
plt.fill_between(years, tv_shows, color="orange", alpha=0.3)

plt.xlabel("Year", fontsize=12)
plt.ylabel("Number of Movies / TV Shows", fontsize=12)
plt.title("Trend")
plt.legend(title="Content Type")

plt.show()
```

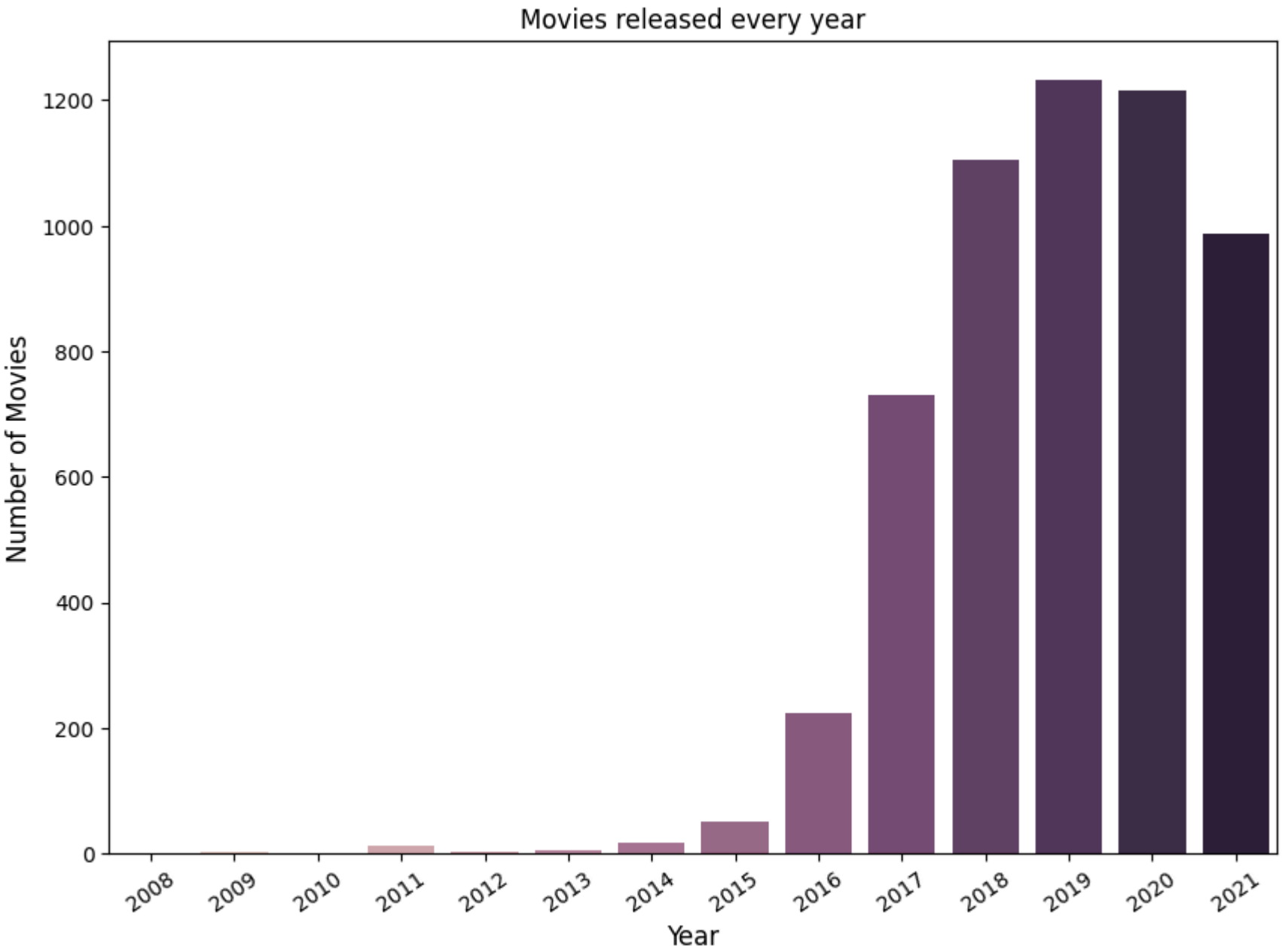


After 2015 Netflix started producing content in great values till 2020 but the count came down in 2020 due to COVID.

In [ ]:
```
#movies released every year

movies_year_df = netflix_df.loc[netflix_df["type"]=="Movie"]["year_added"]
movies_count =movies_year_df.value_counts().reset_index()

plt.figure(figsize=(10,7))
bar_plot = sns.barplot(data=movies_count, x="year_added", y="count", hue="year_added")
plt.xlabel("Year", fontsize=12)
plt.ylabel("Number of Movies", fontsize=12)
plt.title("Movies released every year")
plt.xticks(rotation=35)
bar_plot.legend_.remove()

movies_count.iloc[:10]
```

Out[ ]:

	year_added	count
0	2019	1233
1	2020	1215
2	2018	1106
3	2021	988
4	2017	731
5	2016	224
6	2015	50
7	2014	18
8	2011	13
9	2013	6

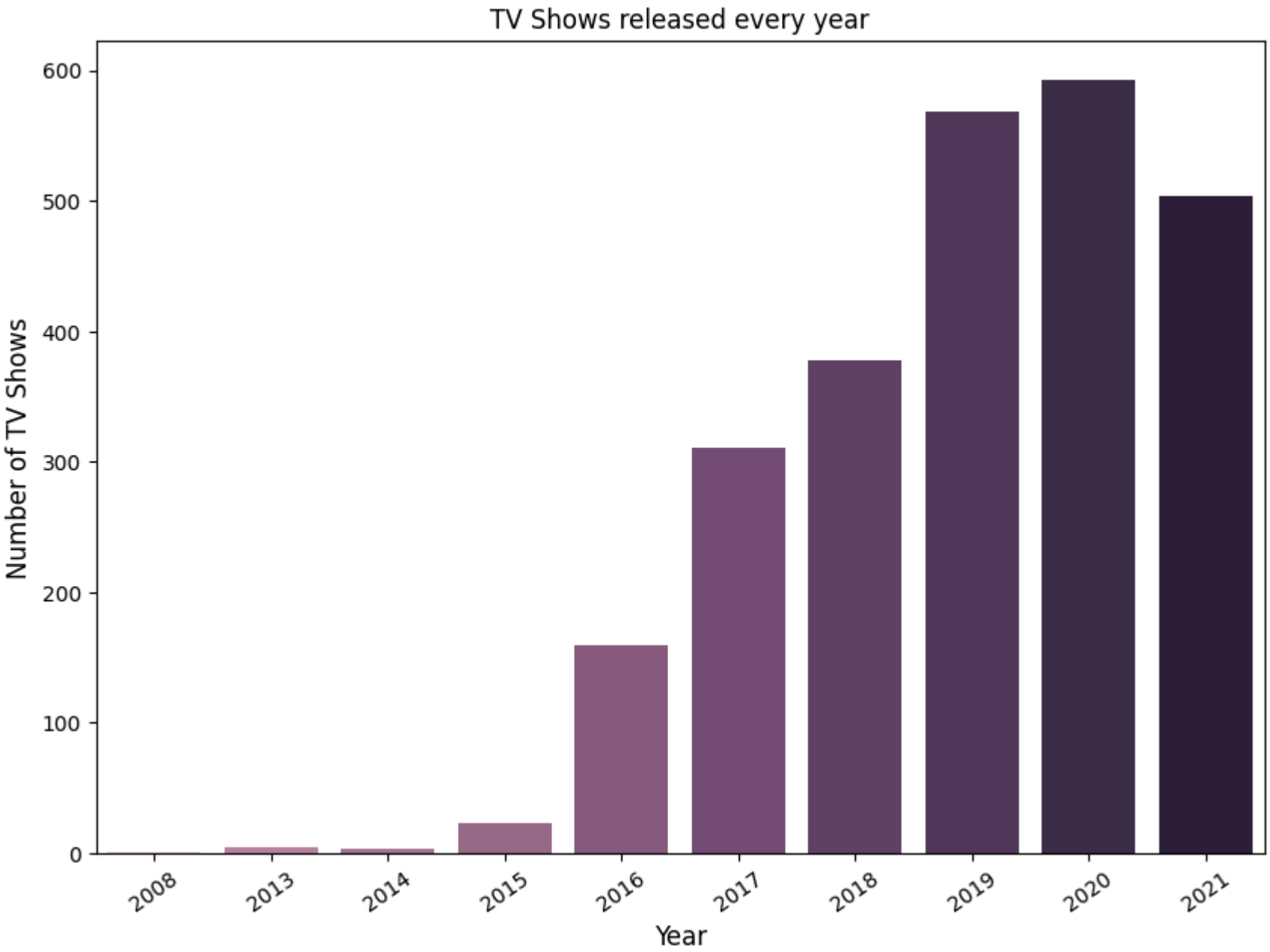


Maximum number of movies were produced in 2019.

In [ ]:
```
#shows released every year
shows_year_df = netflix_df.loc[netflix_df["type"]=="TV Show"]["year_added"]
shows_per_year = shows_year_df.value_counts().reset_index()

plt.figure(figsize=(10,7))
bar_plot = sns.barplot(data=shows_per_year, x="year_added", y="count", hue="year_added")
plt.xlabel("Year", fontsize=12)
plt.ylabel("Number of TV Shows", fontsize=12)
plt.title("TV Shows released every year")
plt.xticks(rotation=35)
bar_plot.legend_.remove()

shows_per_year.iloc[0:10]
```

Out[ ]:

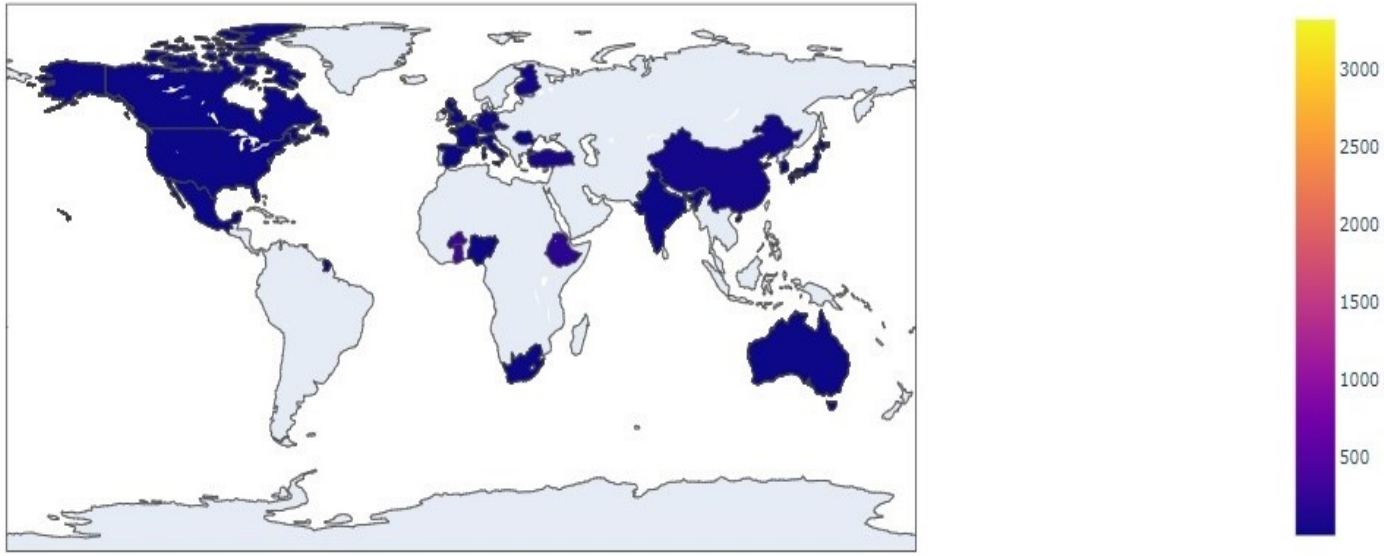
	year_added	count
0	2020	593
1	2019	568
2	2021	504
3	2018	378
4	2017	311
5	2016	159
6	2015	23
7	2013	5
8	2014	4
9	2008	1



Maximum shows were produced in 2020.

In [ ]:
```
import plotly.graph_objects as go
from plotly.offline import init_notebook_mode, iplot
```

In [ ]:
```
#content by countries
filtered_countries = netflix_df.set_index('title').country.str.split(', ',
expand=True).stack().reset_index(level=1, drop=True);
filtered_countries = filtered_countries[filtered_countries != 'Country Unavailable']
iplot([go.Choropleth(
locationmode='country names',
locations=filtered_countries,
z=filtered_countries.value_counts()
)])
```

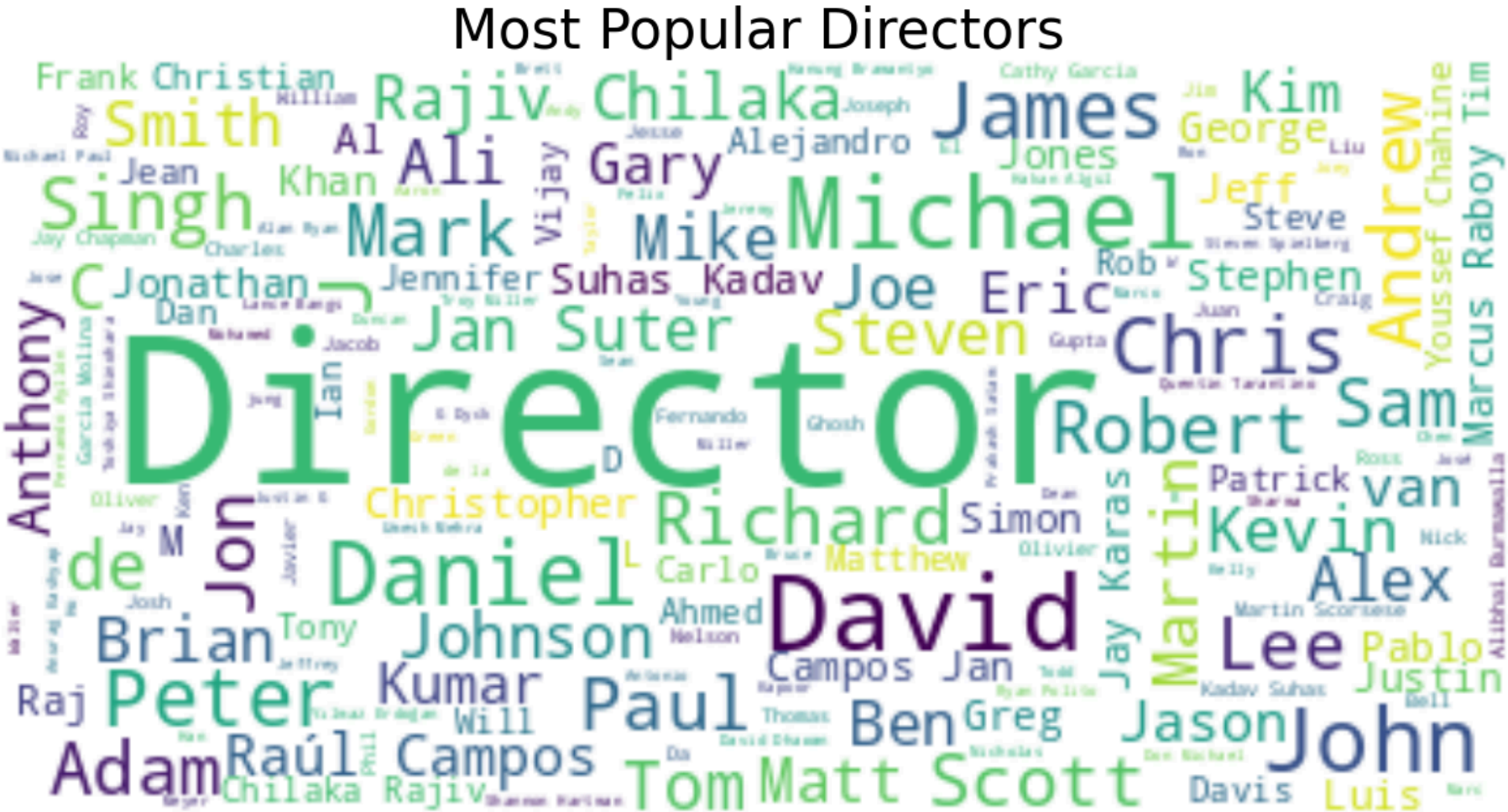


USA produces the maximum content followed by India and United Kingdom.

In [ ]:
```
#top directors on netflix
from wordcloud import WordCloud, ImageColorGenerator
text = " ".join(str(each) for each in netflix_df.director)


wordcloud = WordCloud(max_words=200, background_color="white").generate(text)
```

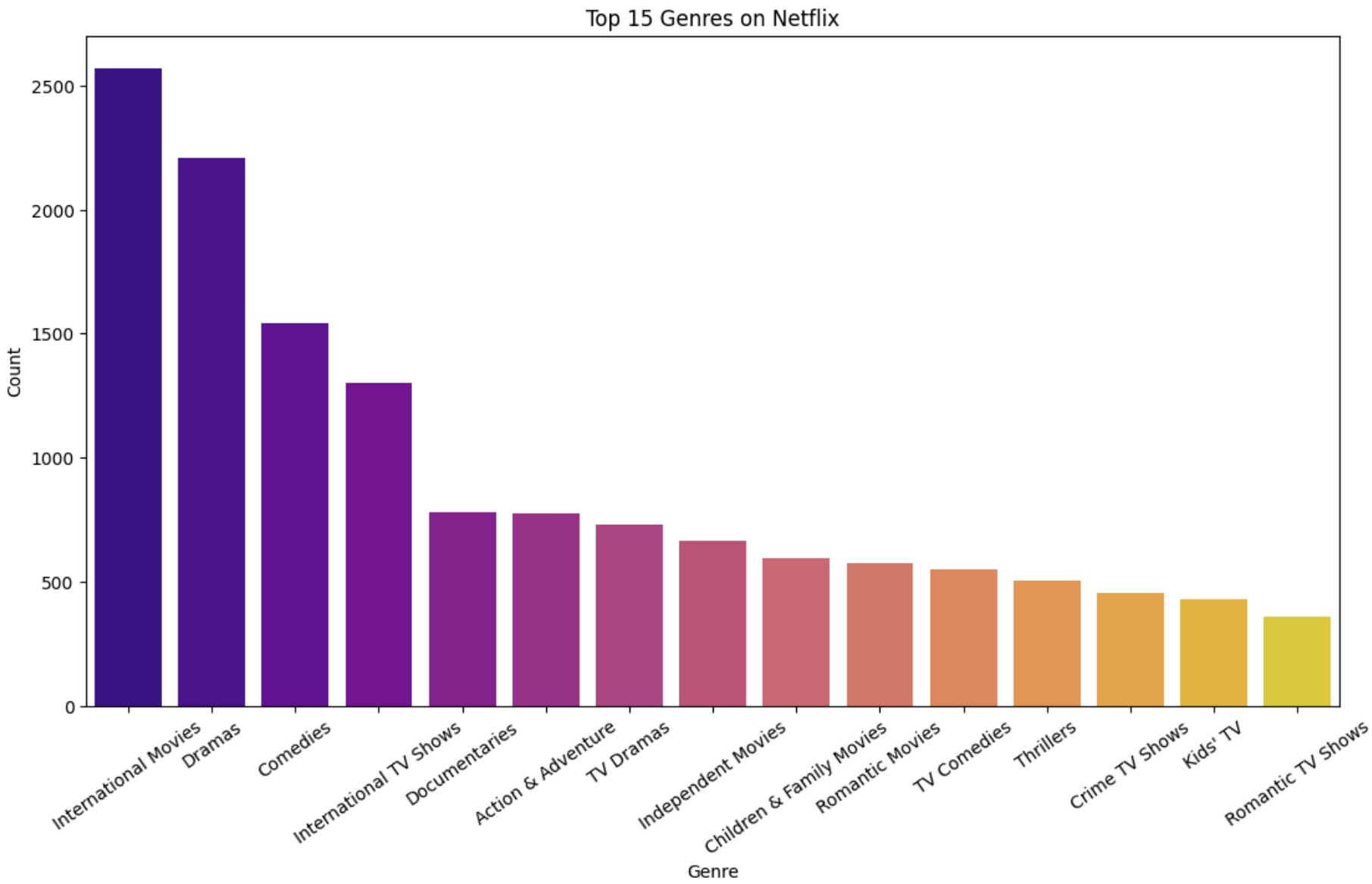
```
plt.figure(figsize=(10,6))
plt.figure(figsize=(15,10))


plt.imshow(wordcloud, interpolation='Bilinear')
plt.title('Most Popular Directors',fontsize = 30)
plt.axis("off")
plt.show()
```

```
<Figure size 1000x600 with 0 Axes>
```



Rajiv Chilaka, Jan Suter, Raul Campos have directed the most number of movies/tv shows.

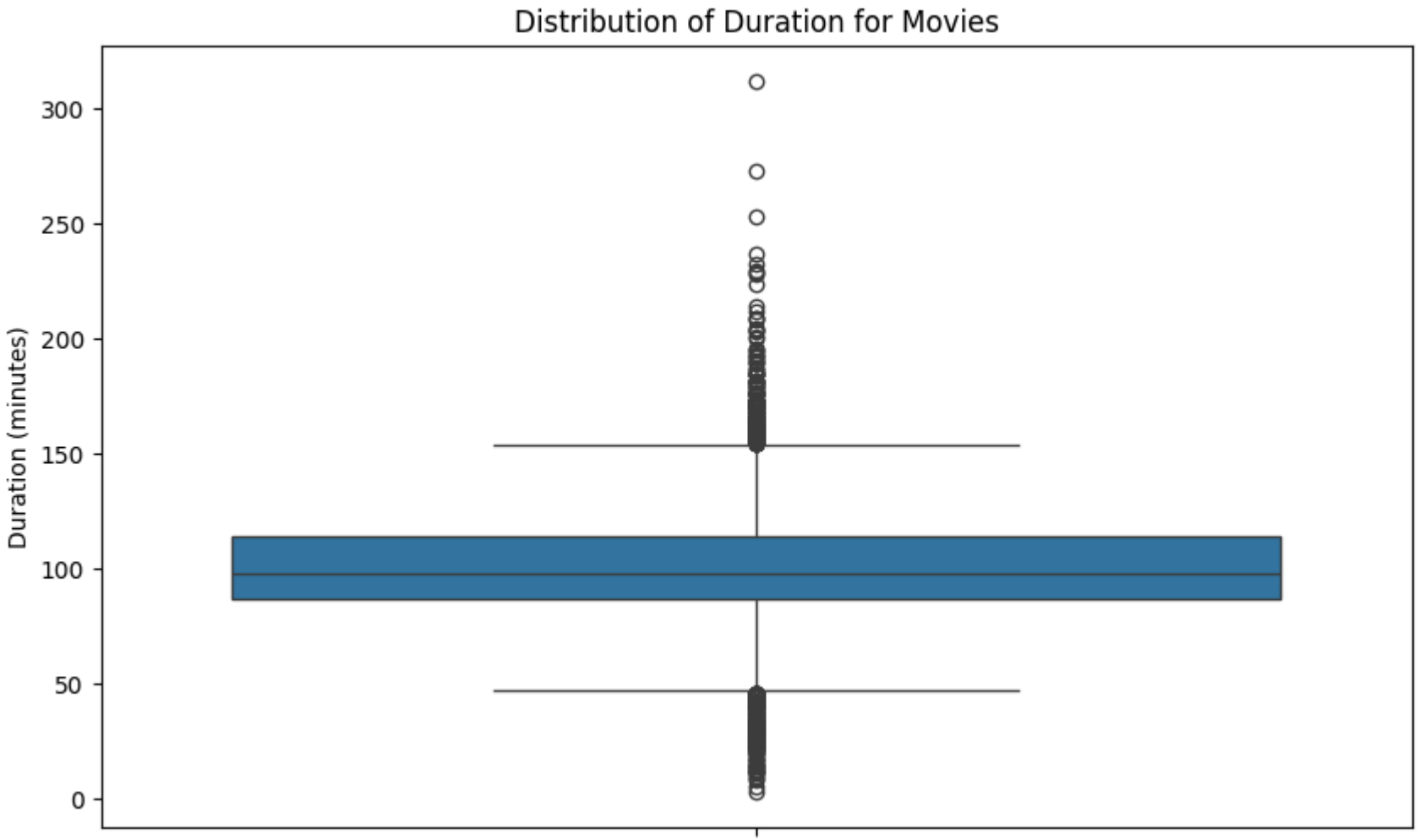
In [ ]:
```
#top 15 genres on netflix
filtered_genres = netflix_df["listed_in"].explode().value_counts().reset_index().iloc[:15]
plt.figure(figsize=(13,7))
g = sns.barplot(data = filtered_genres, x="listed_in", y="count", hue="listed_in", palette="plasma")
plt.title('Top 15 Genres on Netflix')
plt.xlabel('Genre')
plt.ylabel('Count')
plt.xticks(rotation=35)
plt.show()
```



Most popular genres on Netflix are International Movies, Dramas, Comedies, and International TV Shows.

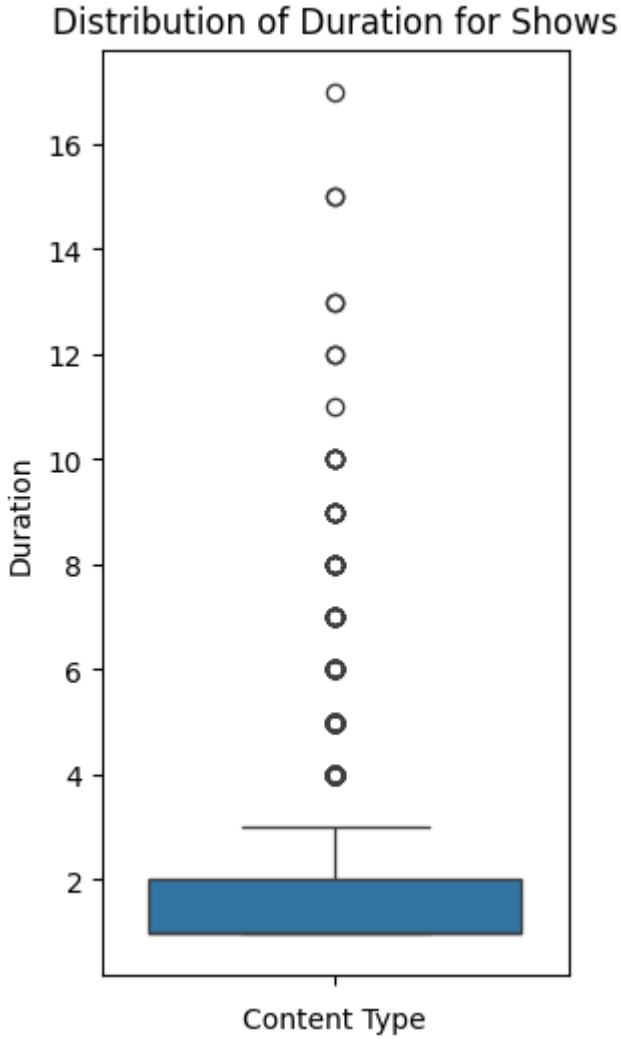
4.2 For categorical variable(s): Boxplot

In [11]:
```
#duration distribution of movies
netflix_movies_df = netflix_df[netflix_df.type.str.contains("Movie")].copy()
netflix_movies_df.loc[:, 'duration'] = pd.to_numeric(
    netflix_movies_df['duration'].str.extract('(\d+)', expand=False), errors='coerce'
)
netflix_movies_df.dropna(subset=['duration'], inplace=True)

plt.figure(figsize=(10, 6))
sns.boxplot(data=netflix_movies_df, y='duration')
plt.ylabel('Duration (minutes)')
plt.title('Distribution of Duration for Movies')
plt.show()
```



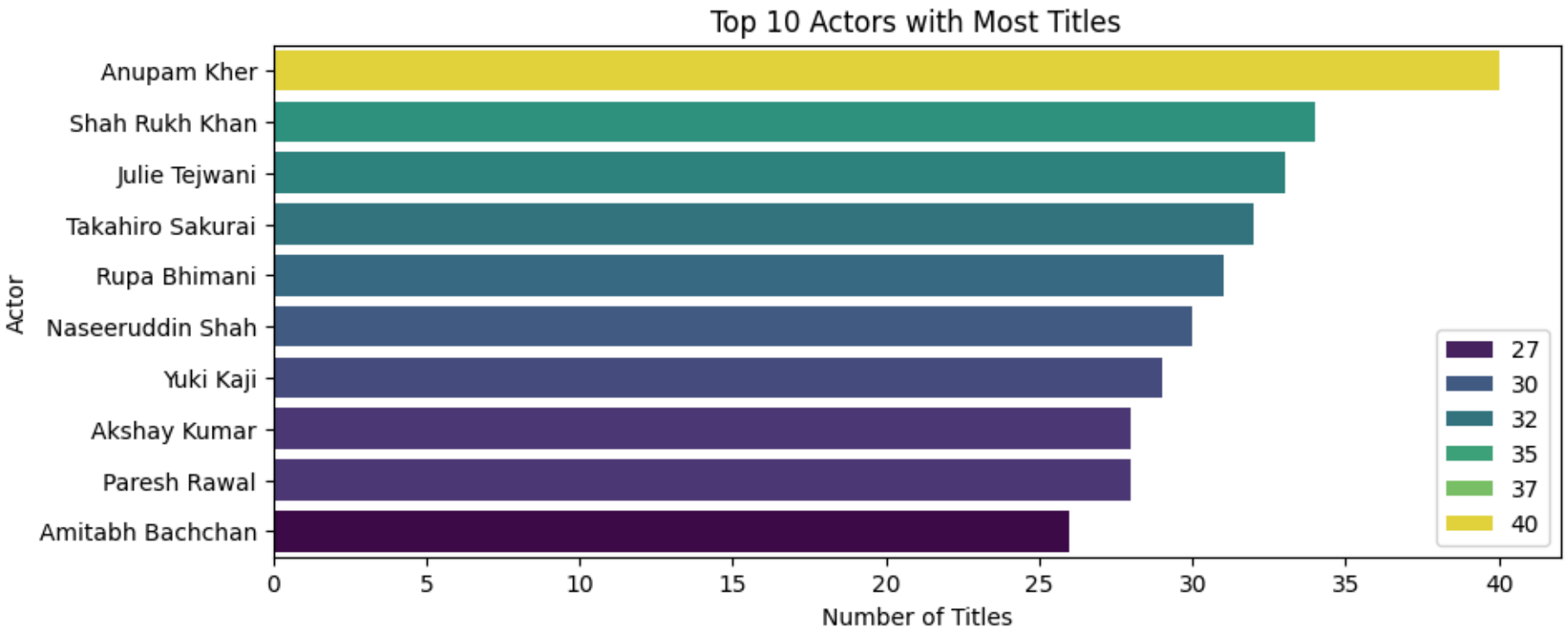
In [12]:
```
#duration distribution of tv shows
netflix_shows_df = netflix_df[netflix_df.type.str.contains("TV Show")].copy()
netflix_shows_df.loc[:, 'duration'] = pd.to_numeric(
    netflix_shows_df['duration'].str.extract('(\d+)', expand=False), errors='coerce'
)
netflix_shows_df.dropna(subset=['duration'], inplace=True)

plt.figure(figsize=(3, 6))
sns.boxplot(data=netflix_shows_df, y='duration')
```

```
plt.xlabel('Content Type')
plt.ylabel('Duration')
plt.title('Distribution of Duration for Shows')
plt.show()
```



In [ ]:
```
#Top 10 actors on netflix
unnested_cast = netflix_df.copy()
# Stripping any leading/trailing whitespace from the cast names
```

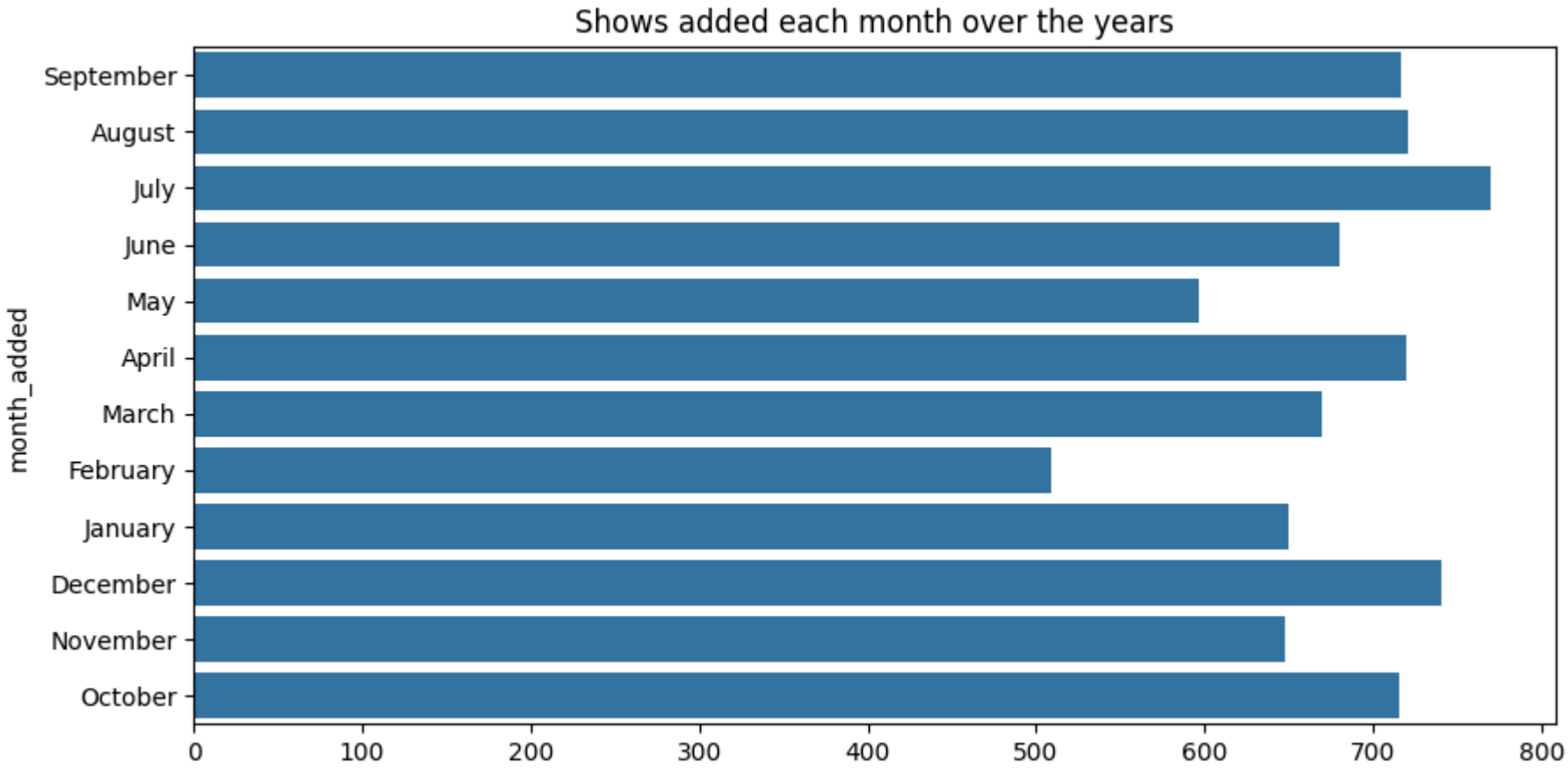
```
unnested_cast['cast'] = netflix_df['cast'].str.strip().str.split(", ")
unnested_cast = unnested_cast.explode('cast')


# Grouping by 'cast' and counting unique 'show_id' (titles)
unique_cast_titles_count = unnested_cast.groupby('cast')['show_id'].nunique().sort_values(ascending=False).iloc[1:11]
unique_cast_titles_count

#Plotting the top 10 actors
plt.figure(figsize=(10, 4))
sns.barplot(y=unique_cast_titles_count.index,x=unique_cast_titles_count.values,hue=unique_cast_titles_count.values,palette='vi
plt.title('Top 10 Actors with Most Titles')
plt.xlabel('Number of Titles')
plt.ylabel('Actor')
plt.show()
```



Anumpam Kher, Shah Rukh Khan and Julie Tejwani are the actors who have done the most work with Netflix.

In [ ]:
```
# Number of shows added on different months
plt.figure(figsize = (10,5))
sns.countplot(netflix_df['month_added'])
plt.title('Shows added each month over the years')
plt.xlabel('')
```

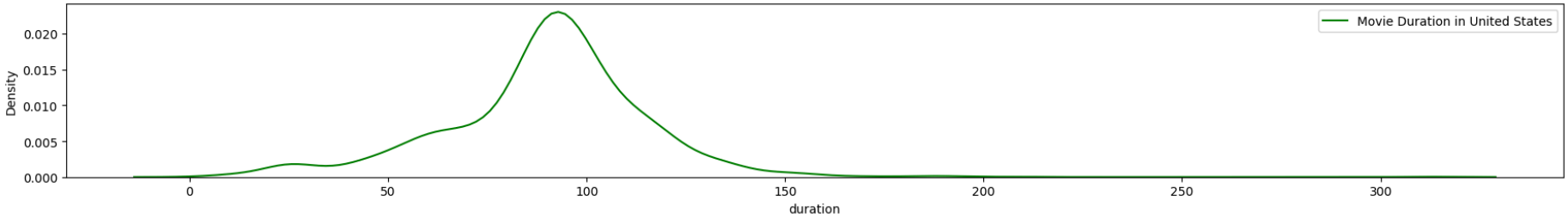
Out[ ]: Text(0.5, 0, '')

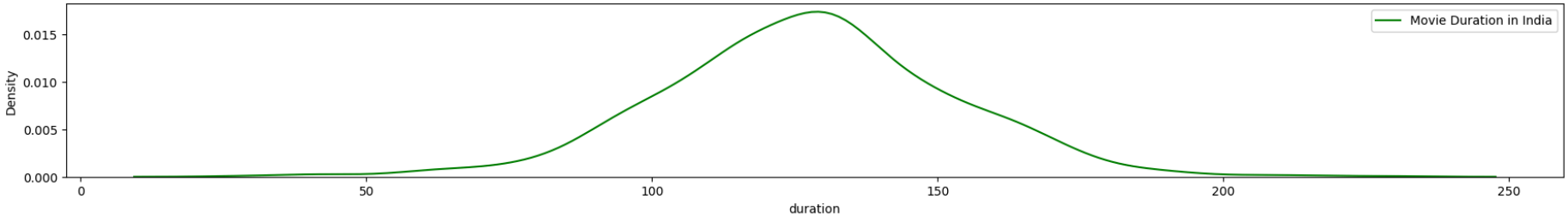


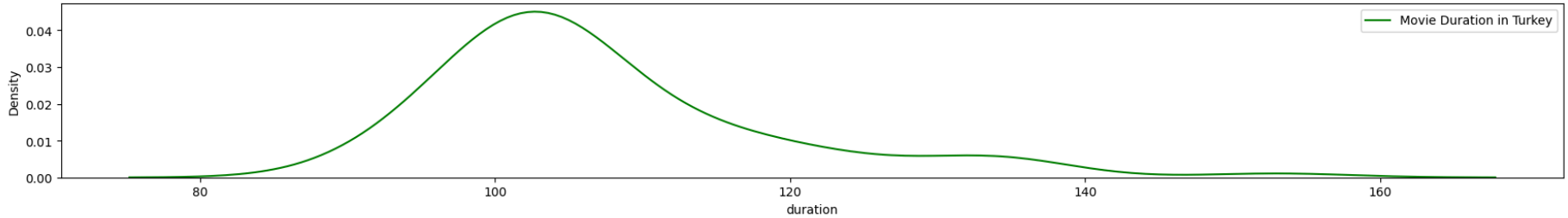
Most shows were added in July and least were added in February.

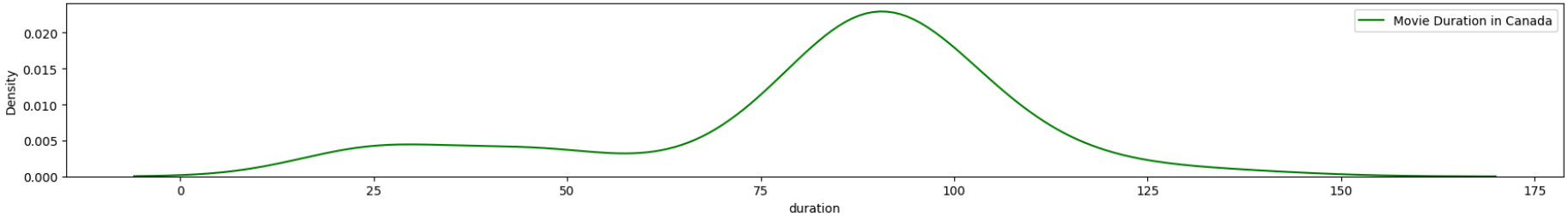
In [ ]:
```
#average movie duration across different countries
countries_0_7 = ['United States', 'India', 'Turkey', 'Canada', 'United Kingdom', 'Spain', 'South Korea']
```

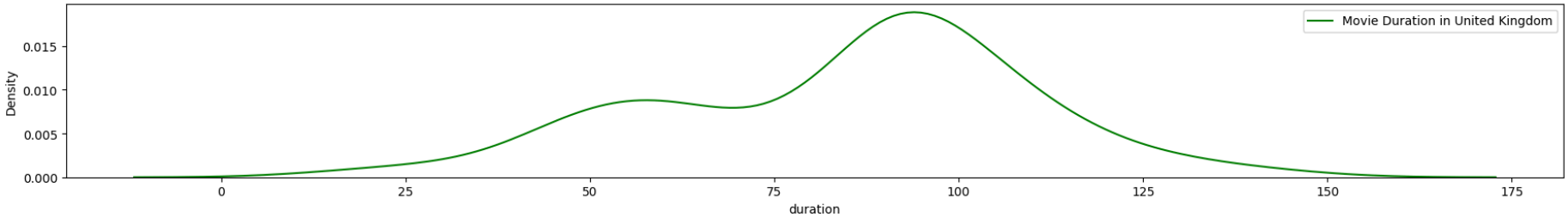
```python
def country_mov_dur(country):
    netflix_country_dur = netflix_df.loc[
        (netflix_df['country'] == country) & (netflix_df['type'] == 'Movie'), 'duration'
    ].dropna()

    # Extract numeric values from duration
    netflix_country_dur = netflix_country_dur.str.replace(' min', '').str.replace(' Season', '', regex=False)
    netflix_country_dur = netflix_country_dur.astype(int)  # Convert to integer

    return netflix_country_dur

# Plot for countries_0_7
fig, axes = plt.subplots(7, 1, figsize=(18, 18))
for i, country in enumerate(countries_0_7):
    country_mov_duration = country_mov_dur(country)
    sns.kdeplot(country_mov_duration, label=f'Movie Duration in {country}', color='g', ax=axes[i])
    axes[i].legend()

plt.tight_layout()
plt.show()
```

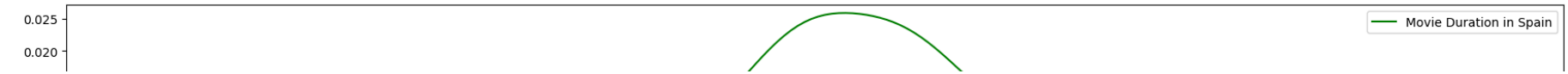

Movie Duration in United States
Density
duration

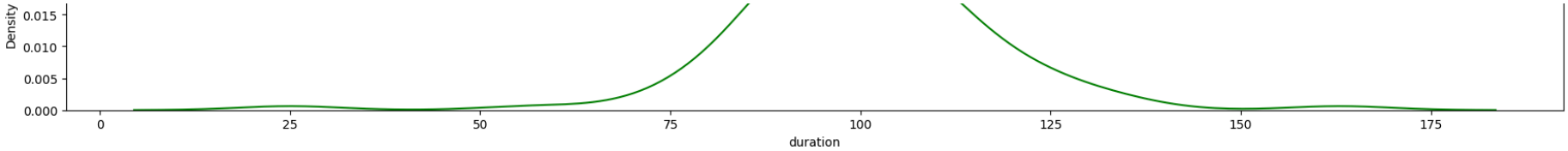

Movie Duration in India
Density
duration

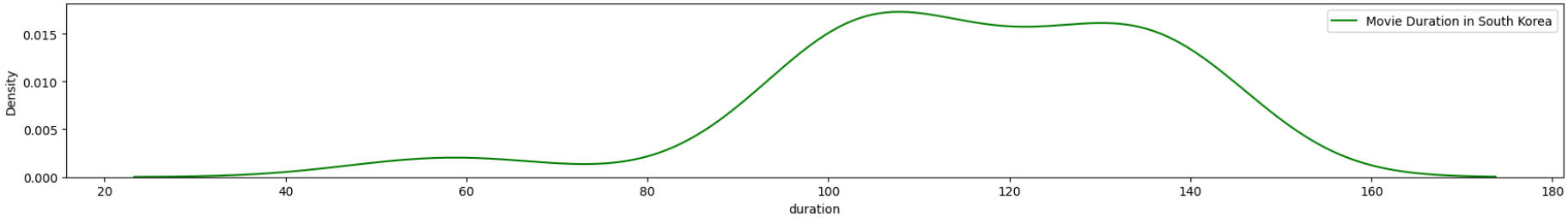

Movie Duration in Turkey
Density
duration


Movie Duration in Canada
Density
duration


Movie Duration in United Kingdom
Density
duration

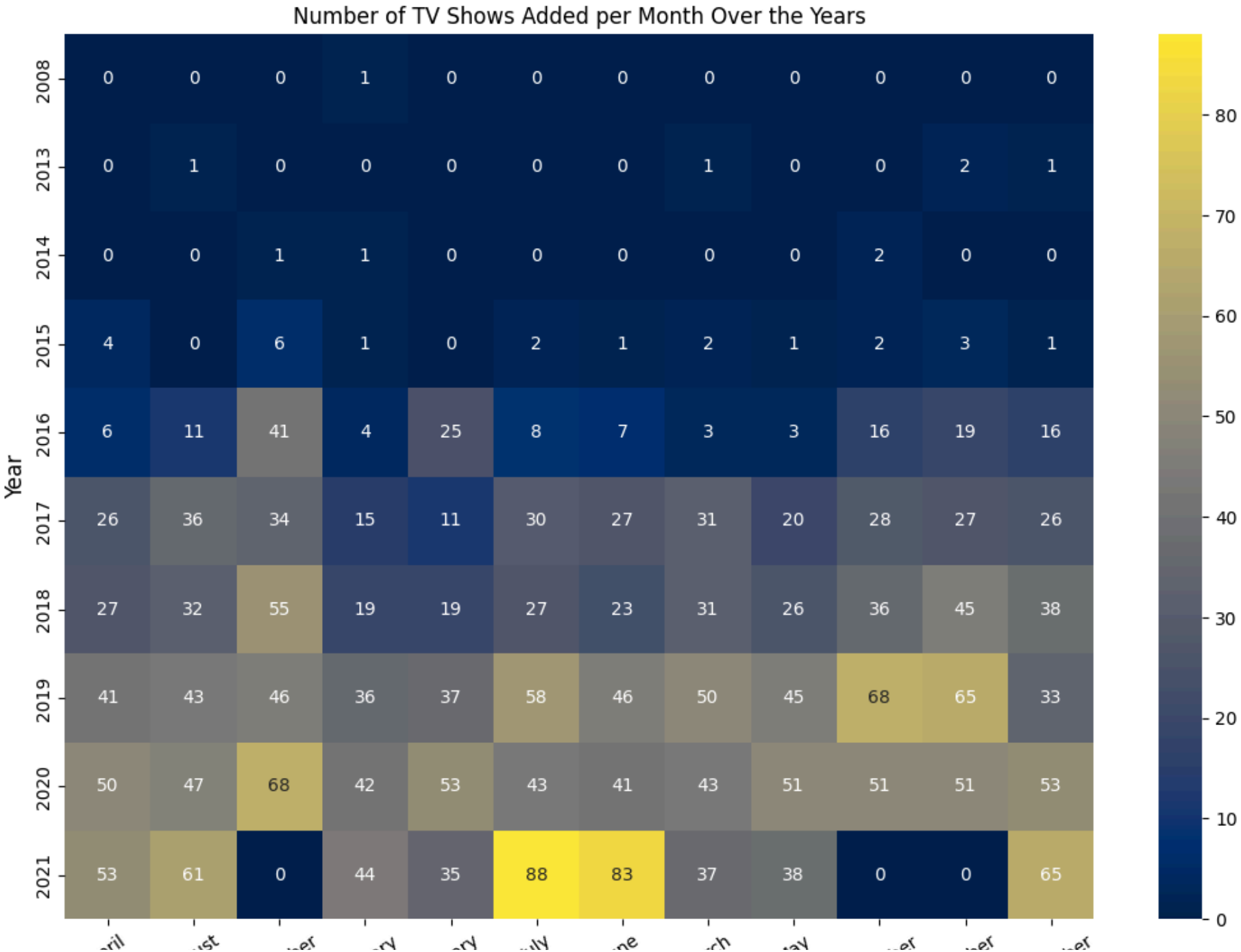

Movie Duration in Spain





4.3 For correlation: Heatmaps, Pairplots

In [ ]:
```
#tv shows added every month over the years
tv_shows = netflix_df[netflix_df["type"]=="TV Show"]
tv_shows_pivot = tv_shows.pivot_table(index="year_added", columns="month_added", aggfunc="size", fill_value=0)

plt.figure(figsize=(13,9))
sns.heatmap(tv_shows_pivot, cmap="cividis", annot=True, fmt="d")
plt.title("Number of TV Shows Added per Month Over the Years")
plt.xlabel("Month", fontsize=12)
plt.ylabel("Year", fontsize=12)
plt.xticks(rotation=35)
plt.show()
```


Number of TV Shows Added per Month Over the Years
Year
2008
2013
2014
2015
2016
2017
2018
2019
2020
2021
0 0 0 1 0 0 0 0 0 0 0 0
0 1 0 0 0 0 0 1 0 0 2 1
0 0 1 1 0 0 0 0 0 2 0 0
4 0 6 1 0 2 1 2 1 2 3 1
6 11 41 4 25 8 7 3 3 16 19 16
26 36 34 15 11 30 27 31 20 28 27 26
27 32 55 19 19 27 23 31 26 36 45 38
41 43 46 36 37 58 46 50 45 68 65 33
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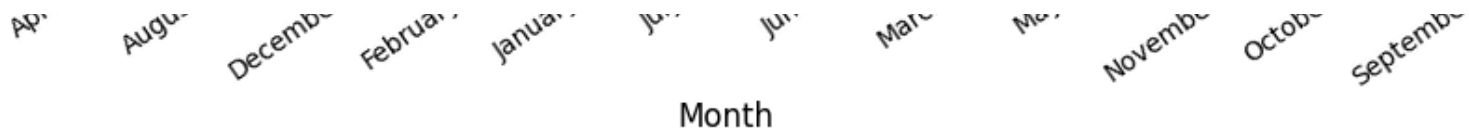
53 61 0 44 35 88 83 37 38 0 0 65
80
70
60
50
40
30
20
10
0



7. Insights

- Netflix has 31.3% shows and 68.7% movies.
- Number of movies/tv shows released on netflix increased drastically after the year 2015 and kept on increasing till 2020 until the world was hit by COVID, a drop in the content release was seen in the year 2020.
- The content produced by USA is far more than any other country. No other country even comes close.
- Most popular directors on Netflix who have directed the most content are Rajiv Chilaka, Jan Suter and Raul Campos.
- Top genres for movies are International Movies, Dramas, and Comedies.
- Top genres for tv shows are International TV Shows, TV Dramas, and TV Comedies.
- Most TV shows on Netflix have one season, suggesting a preference for shorter series among viewers.
- Average duration for mvoies is around 100 minutes.
- Actors who have worked the most with Netflix are Anumpam Kher, Shah Rukh Khan and Julie Tejwani, all 3 belonging to India.
- Most tv shows/movies were added in July, followed by December and the least were added in February.
- The distribution of ratings over time provides information on how the content landscape and audience response are changing. The majority of films are rated TV-MA, which suggests that mature content is preferred.

8. Recommendations

- Some people may prefer watching TV shows than movies, Netflix must concentrate on TV shows as they only contribute around 30% of the total content
- To boost the popularity of new films or television series by hiring the top directors who may help in enhancing the creativity.
- In addition to contacting top directors, Netflix may also approach directors with fewer films and excellent ratings since this can help in reducing the budget but without compromising on the quality. In this case, Netflix can reach a wider audience who may admire the work of a specific director.
- Some specific genres of shows could be given priority so that more number of seasons can be made and keep the audience engaged like thriller.
- Most of the content is being made for the USA audience but could be diversified to attract more and more subscribers.

- The amount of content produced each year might be raised to help customers understand the value they are receiving for the subscription fee they pay. This will not help in retaining the existing customers but could attract new customers too.
- More content could be released during the holiday season to get the maximum watch hours by engaging customers.
- To some extent, the subscription rates also decide the customer retention and onboarding of new customers, so that can also be reduced if necessary by looking at the prices at which the competitors are offering the subscription.
- Since younger audiences use the internet more frequently than older audiences, content created for them could be beneficial.
- Collaborating with top actors and actresses could attract more audience like in India people love movies and tv shows and follow them religiously.
- Prioritizing running ad campaigns in countries where the viewership is low could help in attracting new viewers.